

Final R&D Problem Statement

When a citizen reports a pollution incident in an Indian city, no system today closes the loop between that complaint and a measured result. CPCB's own SAMEER app proves citizens will report incidents at volume - but every report is then routed manually, with no automated diagnosis of what caused it, no estimate of who's affected, and no record of whether the response actually worked. AirSense will build that missing intelligence layer: for a reported incident - starting with construction and road dust in Pune, the source category CPCB's and TERI's own studies already flag as dominant - automatically infer the probable source, estimate the wind-adjusted affected zone, recommend an intervention, and measure whether AQI actually improved afterward. The target user is dual: the citizen who filed the complaint, and the municipal nodal officer who is required to resolve it within 24 hours and currently has no tool to do so.

Here's exactly what changed from your original statement, and why - each change traces back to a specific finding in the competitive analysis, not a stylistic preference:

1. "No system closes the loop" replaced the generic framing - because it's now evidenced, not assumed.

Your original statement was a strong hypothesis. This version states it as a documented fact: SAMEER's 2.4★ rating and the unresolved-complaint reviews are the receipt. That matters for a college proposal - you're no longer asking them to trust your intuition about a gap, you're showing them CPCB's own app store page.

2. Added a specific first vertical: construction/road dust, Pune.

The original was scoped to "a neighbourhood" in the abstract. The blueprint section flagged this as the right narrow starting wedge - visually obvious, high complaint volume, and the one source category where you can validate against wind + PM10 spike + a nearby sensor without needing government data access on day one. Building the general pipeline first, on invented incidents, would have been the classic overreach mistake.

3. Made "measure whether it worked" a co-equal deliverable, not a trailing clause.

It was already in your original sentence, but as the last item in a list it reads as an afterthought. The research shows it's actually the single rarest capability in the entire landscape - nobody, including PRANA at the national level, automates this. It's now positioned as the centerpiece, because it's your actual differentiator.

4. Added the nodal officer as a second, co-equal target user.

Your original only implied the citizen's point of view. The founder's-blueprint analysis flagged the nodal officer - the person CPCB already legally requires to resolve a SAMEER complaint in 24 hours - as arguably the stronger first customer, since they have a deadline and no tool, not just a curiosity about air quality.

5. Deliberately did not add anything about forecasting accuracy, satellite data, or spatial coverage.

Tempting to fold in, but the research is clear that SAFAR, CAMS, and Blue Sky Analytics already win on measurement and forecasting. Scope creep into "also predict AQI better" would drag you back into the most crowded, least defensible lane in the whole landscape.

This is the version I'd take into a build plan or your next college review - it's narrower than your original on purpose, which is what makes it buildable rather than just directionally correct.